Merra2BC User manual

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1 Introduction

The Merra2BC interpolator creates initial and boundary conditions based on MERRA-2 reanalysis (**?**) for a WRF-Chem simulation by interpolating chemical species mixing ratios defined on the MERRA-2 grid to WRF-Chem grid. For the initial conditions, interpolated values are written to each node of the WRF-Chem grid. For the boundary conditions, only boundary nodes are affected.

Merra2BC is written in Python. The utility requires additional modules that need to be installed in the Python environment: NumPy (numpy), NetCDF4 (netcdf4) interface to work with NetCDF files, and SciPy (scipy) package.

The full MERRA-2 reanalysis data set including aerosol and gaseous collections is publicly available online:

- `https://disc.gsfc.nasa.gov/datasets/M2I3NVAER_5.12.4/summary`
- `https://disc.gsfc.nasa.gov/datasets/M2I3NVCHM_5.12.4/summary`

Depending on the requirements, all or one of the following aerosol and gaseous collections need to be downloaded: *inst3_3d_aer_Nv* – gaseous and aerosol mass mixing ratios, ($\mathrm{kg\,kg^{-1}}$) and *inst3_3d_chm_Nv* – carbon monoxide and ozone mass mixing ratios, ($\mathrm{kg\,kg^{-1}}$). Besides downloaded mass mixing ratios, pressure thickness *DELP* and surface pressure *PS* fields also need to be downloaded. Spatial coverage of the MERRA-2 files should include the area of the simulation domain. The time span of the downloaded files should match the start and duration of the simulation. More information regarding MERRA-2 files' specification is provided in (**?**).

1.1 Reconstruction of the pressure in MERRA-2 and in WRF-Chem

Atmospheric pressure is used as a vertical coordinate. Latitude and longitude serve as the horizontal coordinates. The MERRA-2 vertical grid has 72 model layers which are on a terrain-following hybrid $\sigma - p$ coordinate. The pressure at the model top is a fixed constant, $P_{TOP} = 0.01\,\mathrm{hPa}$. Pressure at the model edges is computed by summing the *DELP* starting at P_{TOP}. A representative pressure for the layer can then be obtained by averaging pressure values on

adjacent edges. Indexing for the vertical coordinate is from top to bottom; i.e., the first layer is the top layer of the atmosphere (P_{TOP}), while the 72nd layer is adjacent to the Earth's surface.

In WRF-Chem, the pressure field is not given in *wrfinput_d01* and *wrfbdy_d01* files. Hence, the pressure field must be restored using surface pressure P_{SFC} taken from *met_em_...** files created by *metgrid.exe* during the preprocessing stage. Pressure at the top of the model *wrf_p_top* and η values on half levels (*znu*) are taken from the *wrfinput_d01* file. The procedure of reconstructing the pressure from *met_em_...** files using the Python code is demonstrated in Fig. **??**.

```
def get_pressure_from_metfile(metfile):
    PSFC=metfile.variables['PSFC'][:]
    WRF_Pres = np.zeros([nz,ny,nx])
    for z_level in reversed(range(nz)):
        WRF_Pres[nz-1-z_level,:]=PSFC*znu[0,z_level] +
                        (1.0 - znu[0,z_level])*wrf_p_top
    return WRF_Pres
```

Figure 1: A Python script, which reconstructs the pressure using the *met_em_...** files. nx, ny, and nz indicate the number of grid nodes in WRF-Chem domain.

1.2 Mapping chemical species between MERRA-2 and WRF-Chem

Merra2BC file *src/config.py* contains multiplication factors to convert MERRA-2 mass mixing ratios of gases given in $\mathrm{kg\,kg^{-1}}$ into ppmv. Aerosols are converted from $\mathrm{kg\,kg^{-1}}$ to $\mu\mathrm{g\,kg^{-1}}$. When using the GOCART aerosol module in WRF-Chem simulation, all MERRA-2 aerosols and gases are matched with those from WRF-Chem. We simply multiply by a factor of 10^9 to convert MERRA-2 aerosol mixing ratios given in $\mathrm{kg\,kg^{-1}}$ into $\mu\mathrm{g\,kg^{-1}}$. In the case of gases, we need to multiply MERRA-2 mass mixing ratios by a ratio of molar masses M_{air}/M_{gas} multiplied by 10^6 to convert $\mathrm{kg\,kg^{-1}}$ into ppmv, where M_{gas} and M_{air} are molar masses ($\mathrm{g\,mol^{-1}}$) of the required gas and air ($28.97\,\mathrm{g\,mol^{-1}}$), respectively. If another aerosol module is chosen in WRF-Chem, then different multiplication factors should be used.

1.3 Interpolation procedure

A brief description of the interpolation procedure applied to the initial conditions is presented in Alg. **??**. For boundary conditions, the procedure is similar, except that additional updates of domain boundary tendencies are required and interpolation is performed for each step, where boundary conditions are applied.

Algorithm 1 Interpolation procedure applied to initial conditions.

Pressure reconstruction at each node of the MERRA-2 and WRF-Chem grids.

for each 72 vertical layers in MERRA-2 grid **do**
    Horizontal interpolation of MERRA-2 pressure on WRF-Chem latitude, longitude nodes using bivariate spline approximation (method *RectBivariateSpline* from Scipy module).
Result: MERRA-2 pressure is calculated on 72 levels but on latitude, longitude nodes of the WRF-Chem grid.

for each chemical species mixing ratio **do**
    for each 72 vertical layers in MERRA-2 grid **do**
        Horizontal interpolation of MERRA-2 species mixing ratio on WRF-Chem latitude, longitude nodes using bivariate spline approximation (method *RectBivariateSpline* from Scipy module).
    Result: MERRA-2 species mixing ratio is calculated on 72 levels but on latitude, longitude nodes of WRF-Chem grid.

    for each lat, long node of the WRF-Chem grid **do**
        Vertical linear interpolation of MERRA-2 species mixing ratio on WRF-Chem vertical coordinate (method *interp1d* from from Scipy module).
    Result: MERRA-2 species mixing ratio is interpolated at each node of WRF-Chem grid.

    Multiplying interpolated species mixing ratio by corresponding factor to convert $\mathrm{kg\,kg^{-1}}$ into ppmv or $\mu\mathrm{g\,kg^{-1}}$, depending whether it gas or aerosol.
    Updating corresponding fields in WRF-Chem *wrfinput_d*01 file by interpolated values.
Result: WRF-Chem grid is updated by interpolated values from MERRA-2 grid.

1.4 Typical workflow

Here are the steps describing how to work with the Merra2BC interpolator:

1. Run *real.exe*, which will produce initial *wrfinput_d01* and boundary conditions *wrfbdy_d01* files required by the WRF-Chem simulation.

2. Download required MERRA-2 files from `https://disc.gsfc.nasa.gov/datasets/M2I3NVAER_5.12.4/summary` and `https://disc.gsfc.nasa.gov/datasets/M2I3NVCHM_5.12.4/summary`;

3. Download Merra2BC from `https://github.com/saneku/Merra2BC`.

4. Set defaults in *src/config.py* file, which contains

   a. mapping of chemical species and aerosols between MERRA-2 and WRF-Chem;

   b. paths to *wrfinput_d01*, *wrfbdy_d01*, and *met_em_...** files;

   c. a path to the downloaded MERRA-2 files.

5. Runtime arguments can override configuration values without editing *src/config.py*: `--wrf_dir`, `--wrf_input_file`, `--wrf_bdy_file`, `--wrf_met_dir`, `--wrf_met_files`, `--mera_dir`, `--mera_files`, `--do_IC`/`--no_do_IC`, and `--do_BC`/`--no_do_BC`. Use "python *main.py* –help" to print all options.

6. *real.exe* sets default boundary and initial conditions for some chemical species. Merra2BC adds interpolated values to the existing values, which may cause incorrect concentration values. To avoid this, run "python *zero_fields.py*", which will zero the required fields.

7. Run "python *main.py*", which will do the interpolation. As a result, files *wrfinput_d01* and *wrfbdy_d01* will be updated by interpolated MERRA-2 values. Example: "python *main.py* –wrf_dir /path/to/wrf/run –wrf_met_dir /path/to/wps/run –mera_dir /path/to/merra –do_IC –do_BC".

8. Modify the WRF-Chem *namelist.input* file at section *&chem*: set *have_bcs_chem = .true.* to activate updated boundary conditions and, if needed, *chem_in_opt = 1* to activate updated initial conditions.

9. Run *wrf.exe*.

References

Randles, C. A., da Silva, A. M., Buchard, V., Colarco, P. R., Darmenov, A., Govindaraju, R., Smirnov, A., Holben, B., Ferrare, R., Hair, J., and others. (2017). The MERRA-2 aerosol reanalysis, 1980 onward. Part I: System description and data assimilation evaluation. *Journal of Climate*, 30(17), 6823–6850.

Bosilovich, M. G., Lucchesi, R., and Suarez, M. (2016). MERRA-2: File specification GMAO Office Note No. 9 (Version 1.1).